Robin Macharg

Senior iOS Developer

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LinkedIn | GitHub | StackOverflow

Professional Summary

Expert Native iOS Engineering: *Extensive experience building, optimizing, and shipping robust iOS applications using Swift, SwiftUI, and Objective-C to millions of users globally.*

High-Security & Secure Systems: *Proven track record delivering secure mobile architectures for regulated environments, including a VPN-secured client for the UK Government and enterprise-grade penetration-tested applications.*

Modern Architecture & Core SDKs: *Deep expertise in scalable patterns like MVVM, Combine, TCA, and Clean Architecture, with hands-on experience developing core mobile SDKs at scale for platforms like Sky and Living Map.*

Full-Lifecycle App Delivery: *Complete ownership of features end-to-end, from greenfield prototyping and automated App Store/ TestFlight deployment to high-performance maintenance and optimization.*

Collaborative Agile Engineering: *Certified Scrum Master with a history of mentoring engineering teams, optimizing development workflows, and promoting high engineering standards in international teams.*

Security, Performance, and Native iOS Engineering Match

- **Native iOS & Shipping Success:** Shipped multiple high-profile apps and SDKs to the Apple App Store, utilizing both Swift (SwiftUI, Combine, async/await) and Objective-C across enterprise and consumer domains.
- **Expert Tooling & Automation:** Proficient with Xcode, TestFlight, and App Store Connect, with a history of overhauling release procedures and setting up robust Fastlane and GitHub Actions CI/CD pipelines.
- **Scalable Architecture & Design:** Deep practical knowledge of modern design patterns including MVVM, TCA, Redux, and Clean Architecture, ensuring robust, testable, and highly maintainable codebases.
- **Security-First Development:** Handled secure mobile communication, VPN implementations, and enterprise MDM deployments for UK Government and major corporate clients, with an active focus on performance and optimization.
- **Collaborative Agile Delivery:** Certified Scrum Master experienced in cross-functional Agile workflows, git-based collaboration, code reviews, and mentoring team members to drive feature ownership and engineering excellence.

Education, Certifications & Credentials

- **BSc (Hons) Artificial Intelligence and Computer Science (Joint)** — University of Edinburgh
- **Certified Scrum Master (CSM)** — Scrum Alliance

Selected Key Roles

Senior Developer | Floww Ltd.

May 2023 - September 2023

Keywords: [iOS, SwiftUI, TCA, Redux, CI/CD, Shell Scripting]

- Developed core features for a private investment market platform using modern SwiftUI and Composable Architecture (TCA).
- Promoted engineering best practices, introducing componentization, git push lint hooks, and compile-time monitoring to optimize build and startup performance.
- Maintained automated CI/CD pipelines via GitHub Actions and mentored junior team members.

Lead Mobile Developer | This One Ltd.

May 2022 - May 2023

Keywords: [SwiftUI, Combine, Async/Await, Fastlane, Python, GCP]

- Led greenfield-to-seed-round development of an ML-based Movie Recommender app using SwiftUI, Combine, and async/await.
- Managed automated App Store and TestFlight deployment pipelines using Fastlane, streamlining product discovery and rapid releases.
- Built mock backend servers in Python Flask and implemented proof-of-concept hybrid implementations across React Native and Ionic.

Senior iOS Developer | Sky

October 2021 - April 2022

Keywords: [Swift, Kotlin Multiplatform, SwiftUI, Video Streaming, SDKs, Agile]

- Maintained a hybrid Swift and Kotlin Multiplatform Mobile (KMM) Core Video SDK supporting over 25 million active daily users.
- Optimized SDK performance and rewrote the internal video streaming reference application in SwiftUI.
- Integrated streaming delivery technologies and analytics platforms, managing bi-weekly SDK release cycles.

Senior Apple Engineer | Bugsnag Inc.

January 2020 - June 2020

Keywords: [Objective-C, C++, iOS, macOS, Crash Reporting, Open Source]

- Maintained open-source Bugsnag crash-reporting libraries for Apple platforms (iOS, macOS, watchOS, tvOS) in Objective-C and C++.
- Handled complex 3rd-line support issues, specializing in low-level crash log analysis, stack trace inspection, and root-cause isolation.
- Managed package distribution across SPM, CocoaPods, and Carthage with Travis and Buildkite automated testing.

Principal iOS Developer | Amiosec Ltd. (Contract)

March 2019 - December 2019

Keywords: [Xamarin, C#, Objective-C, Swift Interop, Security, EWS]

- Architected and built a custom VPN-secured Exchange email client for the UK Government's Foxhound secure communications project.
- Developed low-level cross-language interop between Xamarin/C# and Objective-C/Swift using .NET embedding.
- Reverse-engineered Exchange Web Services (EWS) via SSL Man-in-the-Middle proxying to ensure strict security and protocol compliance.

Earlier Career Highlights

1996 - 2021 Engineering roles covering iOS SDK development, enterprise security, hardware-level software, middleware, and network analytics.

- **Senior Developer** | *Living Map Ltd.* (iOS, Mapping SDK, XCFrameworks, CocoaPods, Carthage)
- **Principal iOS Developer (Contract)** | *Dyson Ltd.* (iOS, Swift, ARKit, RealmDB, AirWatch MDM, Unit Testing)
- **Head of Mobile Development** | *Actual Experience Ltd.* (Objective-C, Swift, CoreData, Low-Level Networking, Jenkins)
- **Principal Mobile Developer (Contract)** | *Accident Exchange Group* (iOS, CoreLocation, MapKit, CoreData, RestKit)
- **Senior Middleware & DevOps Developer** | *MySpace* (Java, Python, JBoss, Drools, Puppet, Objective-C)
- **Senior Developer** | *RMS Services Ltd.* (Python, PostgreSQL, RESTful APIs, Network Discovery)
- **Aerodynamics & Mass Properties Engineer** | *Airbus UK Ltd.* (Python, PyQt, Perl, PDL, OpenGL)
- **Server & Embedded Developer** | *Sift / Appliance Studio / Obtree / Circle.com* (Linux, Embedded Java, Perl, C, Web Infrastructure)

Open Source Contributions

- Bricksmith: SourceForge award-winning LEGO CAD software for macOS built with Objective-C and C.
- React Native Input Spinner: Contributor to open-source UI component library.
- Pycel / Excel Formula Parser: Python-based Excel dependency graph and formula parser engine.
- Coding Exercises: Solutions and performance-optimized interview algorithms in Swift and Python.

Technical Skills and Expertise

iOS & Apple Platforms - Swift, Objective-C, SwiftUI, Combine, async/await, UIKit, CoreData, RealmDB, XCFrameworks

Tooling & Automation - Xcode, TestFlight, App Store Connect, Fastlane, SPM, CocoaPods, Carthage, GitHub Actions, Travis CI

Security & Core Systems - VPN Integration, AirWatch MDM, Penetration Testing, Low-Level Networking (TCP/ICMP), C/C++, C#, Kotlin Multiplatform (KMM)

Backend & Scripting - Python (FastAPI, Flask), PostgreSQL, Docker, Shell Scripting, RESTful APIs, Git